Compositional Robustness Verification of Heterogeneous Mixture-of-Experts Architectures

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Abstract

Heterogeneous Mixture-of-Experts (MoE) architectures, where domain-specific experts with distinct training data or architectures are coordinated by a learned router, offer modularity and scalability, yet the interaction between routing stability and expert robustness remains underexplored. Verifying full MoE systems is intractable: dynamic routing induces combinatorial behavior, and model scale overwhelms current verifier capacity. We present a **compositional verification framework** that decomposes this problem: if the router provably maintains consistent expert selection within an adversarial perturbation region, and the selected expert is provably robust, then the entire MoE is provably robust—reducing verification from the full system to independent components (20× parameter reduction). Empirically, we show that expert robustness dominates system behavior (13.1 percentage point improvement with adversarial training) while router training has negligible impact (<0.1%) for dissimilar domains. Verified routers achieve 100% certified accuracy at practical threat models. These results establish a principled path toward scalable, formally verified modular AI.

1. Introduction

Heterogeneous Mixture-of-Experts (MoE) models combine domain-specific experts, each potentially with different architectures or training data, coordinated by a learned router. This modularity enables scalable capacity, but conditional execution yields input-dependent architectures with intertwined discrete and continuous behavior. Formal neural network verification, which provides sound guarantees over all inputs in a specified set, already faces scalability challenges for standard networks. MoE architectures further induce a combinatorial explosion from expert selection and routing boundaries. Existing verifiers either fail to scale or rely on coarse relaxations that produce vacuous guarantees. We address this challenge by exploiting the structured partitioning of the input space induced by MoE gating, enabling sound and scalable verification through explicit reasoning over routing regions and expert-level decoupling.

Formal verification of neural networks provides provable robustness guarantees for safety-critical deployment. However, verifying MoE systems poses unique challenges: dynamic routing creates input-dependent computation paths, and the scale of standard MoEs exceeds the capacity of current complete verifiers like α,β-CROWN. Complete neural network verification is NP-complete (Katz et al., 2017), with complexity scaling exponentially in network size. While state-of-the-art verifiers like α,β-CROWN can handle networks up to a few million parameters on specialized benchmarks (Brix et al., 2024), typical MoE systems with hundreds of millions to billions of parameters remain fundamentally intractable. To make verification tractable, we treat routing and expert prediction as separable subproblems and show that their guarantees compose cleanly.

We address this challenge through a **compositional verification** approach. Our central result is:

> **Compositional Robustness Theorem:** *For top-1 routing MoE, if the router maintains invariant expert selection within an ℓ_∞ perturbation ball, and the selected expert is robust within that ball, then the MoE system is robust.*

This decomposition is the key insight: rather than verifying the intractable full system, we verify components independently and compose guarantees. The practical impact is a **20× reduction** in verification complexity (2M → 96K parameters in our experiments).

Unlike homogeneous MoEs where experts share structure and training distribution, heterogeneous systems require the router to perform *domain identification*, a distinct task from within-domain classification. This separation is key to our compositional approach: domain identification typically

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produces large decision margins, enabling tractable router verification.

Contributions.

1. **Compositional Verification Framework for Heterogeneous MoE: 20$\times$ verification speedup** is achieved by formalizing conditions under which heterogeneous MoE robustness decomposes into router stability and expert robustness, enabling incremental expert integration without re-verifying the full system.
2. **Expert Dominance Finding: 13.1 percentage point** adversarial accuracy improvement stems from expert training, which determines system robustness; router adversarial training has negligible impact (<0.1%) for visually dissimilar domains, enabling **4$\times$ training speedup**.
3. **Verification-Aware Design:** We identify architectural requirements for tractable heterogeneous MoE verification and demonstrate that adversarial training is a *prerequisite* for formal verifiability.

2. Background and Related Work

Mixture-of-Experts. The MoE paradigm (Jacobs et al., 1991) decomposes problems into specialized subtasks via gating networks. Shazeer et al. (2017) scaled this through sparse top-k routing, and Riquelme et al. (2021) extended MoE to vision. Recent work explores heterogeneous compositions (Fedus et al., 2022) and modular architectures (Pfeiffer et al., 2023), but formal robustness properties remain unaddressed.

Neural Network Verification. Complete verifiers like α,β-CROWN (Wang et al., 2021) provide sound robustness certificates via bound propagation and branch-and-bound. Certified training methods (CROWN-IBP (Zhang et al., 2020), SABR (Müller et al., 2023)) improve verifiability by training with formal bounds. Randomized smoothing (Cohen et al., 2019) offers probabilistic certificates for larger models. However, all approaches face scalability limits with multi-million parameter systems.

Compositional Verification. Compositional approaches decompose verification into manageable subproblems. Classical control theory provides foundations through passivity, dissipativity, and small-gain theorems for interconnected systems (Willems, 1972; Zames, 1966), while bisimulation relations enable behavioral abstraction in formal methods (Milner, 1989). In robotics and hybrid systems, assume-guarantee reasoning enables modular certificates (Henzinger et al., 1998). For neural networks, modular verification remains nascent. MoEs are uniquely suited to compositional verification because the router explicitly partitions inputs into disjoint expert domains—if this partitioning is stable under perturbation, component guarantees compose directly.

Research Gap. Prior work addresses homogeneous MoE scalability or monolithic neural network verification separately. We provide the first compositional verification framework specifically for *heterogeneous* MoE systems, where domain separation enables tractable router verification and expert-level decomposition bridges modular architecture design with formal robustness guarantees.

3. Compositional Verification Framework

3.1. Problem Setup

Definitions. We distinguish three key properties:

- **Robustness:** A model is ε-robust at input x if its prediction remains correct for all perturbations $x' \in B_\varepsilon(x)$. We measure this via *certified accuracy* (formally verified) and *adversarial accuracy* (empirical, against PGD attacks).
- **Router Stability:** The router is stable at x if expert selection remains unchanged for all $x' \in B_\varepsilon(x)$, i.e., $\arg\max_i \mathcal{G}_i(x') = \arg\max_i \mathcal{G}_i(x)$.
- **Certified vs. Adversarial Accuracy:** Certified accuracy counts inputs with formal robustness proofs; adversarial accuracy counts inputs where heuristic attacks (e.g., PGD) fail. Certified accuracy $\leq$ adversarial accuracy, as certification is strictly stronger.

We also distinguish training regimes: **Robust Training (RT)** uses adversarial examples during training (Madry et al., 2018); **Non-Robust Training (NRT)** uses standard supervised learning without adversarial augmentation.

Notation. Let $x \in \mathbb{R}^d$ be an input with class label $y \in \{1, \ldots, C\}$ where $C = \sum_{i=1}^{N} C_i$ is the total number of classes across all expert domains. We define the ℓ_∞ **perturbation ball** $B_\varepsilon(x) = \{x' : \|x' - x\|_\infty \leq \varepsilon\}$, where $\|z\|_\infty = \max_i |z_i|$ bounds the maximum per-pixel change. For images with pixel values in $[0, 1]$, $\varepsilon = 8/255 \approx 0.031$ represents imperceptible perturbations.

A heterogeneous MoE consists of N domain-specific experts $\{E_i : \mathbb{R}^d \to \mathbb{R}^{C_i}\}_{i=1}^{N}$ and a router $\mathcal{G} : \mathbb{R}^d \to \mathbb{R}^N$ selecting expert $m^* = \arg\max_i \mathcal{G}_i(x)$.

3.2. Main Result

We now formalize the compositional robustness guarantee.

Theorem 3.1 (Compositional Robustness)**.** *The robustness of a heterogeneous MoE system decomposes into two in-*

dependently verifiable conditions: router stability (expert selection invariance) and expert robustness (correct classification within the perturbation region).

Formal Statement. *Let* $\mathrm{MoE} = (\mathcal{G}, \{E_i\})$ *use top-1 routing and let* $m^* = \arg\max_i \mathcal{G}_i(x)$. *If*

$$\forall x' \in B_\varepsilon(x) : \arg\max_i \mathcal{G}_i(x') = m^* \textit{ (stability)} \quad (1)$$

and

$$\forall x' \in B_\varepsilon(x) : \arg\max E_{m^*}(x') = y \textit{ (robustness),} \quad (2)$$

then the MoE is ε*-robust at* x.

Intuition. The router partitions the input space into disjoint expert regions. If these partitions are *stable*—meaning the router satisfies $\arg\max_i \mathcal{G}_i(x') = m^*$ for all $x' \in B_\varepsilon(x)$, so no adversarial perturbation can alter expert selection—and the selected expert maintains correct classification throughout $B_\varepsilon(x)$, then the composed system inherits robustness without requiring monolithic verification.

Proof Sketch. Let $x' \in B_\varepsilon(x)$ be an arbitrary perturbation. By the stability condition (1), the router selects expert m^* for all inputs in $B_\varepsilon(x)$, hence $\mathrm{MoE}(x') = E_{m^*}(x')$. By the robustness condition (2), expert E_{m^*} correctly classifies all such perturbations: $\arg\max E_{m^*}(x') = y$. Since the MoE output is determined solely by the selected expert, we have $\arg\max \mathrm{MoE}(x') = y$ for all $x' \in B_\varepsilon(x)$, establishing ε-robustness. Full proof in Appendix A. □

Remark 3.2 (Top-1 Requirement). Theorem 3.1 requires top-1 routing. For top-k routing ($k > 1$), the weighted combination of multiple experts introduces additional attack surface: adversaries could exploit the weighting mechanism even if individual experts are robust. Extending compositional guarantees to top-k requires verifying both expert selection stability *and* weight stability—a stronger condition we leave to future work.

3.3. Verification Implementation

We use α,β-CROWN for component verification:

Router Verification: Router stability (1) requires that expert m^* remains the argmax throughout $B_\varepsilon(x)$. This holds if and only if $\mathcal{G}_{m^*}(x') > \mathcal{G}_j(x')$ for all competing experts $j \neq m^*$ and all $x' \in B_\varepsilon(x)$. We verify this by checking $\mathcal{G}_{m^*}(x') - \mathcal{G}_j(x') > 0$, reducing stability to $N - 1$ binary properties per input. Expert verification follows the standard formulation (2).

Challenges: Router verification difficulty depends on decision margin geometry. When domains are visually dissimilar (e.g., CIFAR-10 vs MNIST), routers learn large margins ($\gg \varepsilon$), enabling easy certification. Similar domains produce smaller margins and harder verification (see Section 5).

	Certified Routing Acc. (%)		
Training	ε=2/255	ε=4/255	ε=8/255
Non-Robust (NRT)	100	100	60
Robust (RT)	100	100	100

Table 1. Router certified accuracy (40 samples, 5 runs). Both regimes achieve 100% at $\varepsilon \leq 4/255$; RT extends to $\varepsilon = 8/255$.

		Certified Acc. (%)		
Expert	**Train**	ε=2/255	ε=4/255	ε=8/255
CIFAR-10	NRT	0	0	0
	RT	90	54	0
MNIST	NRT	75	0	0
	RT	100	99	95

Table 2. Expert certified accuracy (20 samples each, 5 runs). RT is required; NRT experts are largely unverifiable.

3.4. Verification-Aware Architecture

Both router and experts use **VerifiableCNN** (96K parameters) designed for α,β-CROWN: average pooling (linear, unlike max pooling which introduces additional branching), no BatchNorm (folded into weights), and minimal operators. Training uses PGD adversarial training ($\varepsilon = 8/255$) for robust variants. Full specifications in Appendix B.

4. Experiments

We evaluate on CIFAR-10 and MNIST because their feature manifolds are nearly disjoint, a setting where router margins are large—allowing us to isolate the contribution of expert robustness. Extended experiments on similar domains appear in Appendix C.

4.1. Router Verification Enables Compositional Guarantees

Table 1 shows that both training regimes achieve 100% certified routing accuracy at practical threat models ($\varepsilon \leq 4/255$). This high certification rate occurs because CIFAR-10 and MNIST lie on well-separated manifolds, yielding decision margins far exceeding ε. RT extends certification to larger perturbations ($\varepsilon = 8/255$).

4.2. Expert Verification Requires Robust Training

Table 2 demonstrates that RT is a *prerequisite* for verifiability. NRT experts are largely unverifiable (CIFAR-10: 0% across all ε), while RT enables certification (CIFAR-10: 90% at $\varepsilon = 2/255$; MNIST: 95% at $\varepsilon = 8/255$). Non-robust models produce irregular decision boundaries that lead to loose CROWN bounds.

Configuration	Clean (%)	Adv (%)
Both Experts RT	87.98	**41.93**
Both Experts NRT	**91.32**	28.78
Δ from expert config	*3.34*	*13.15*
Δ from router config	*<0.1*	*<0.1*

Table 3. MoE system accuracy (8 configs, 5 runs). Expert training dominates; router training is negligible.

4.3. Expert Robustness Dominates System Behavior

Table 3 shows that expert training dominates system robustness: **13.15 percentage point** adversarial accuracy improvement from RT experts, while router training contributes $<0.1\%$. All configurations achieve 100% routing accuracy, validating the compositional property. **Practical implication: 4× training speedup** is achieved by omitting router adversarial training for dissimilar domains.

4.4. Scalability

20× parameter reduction (2M full MoE → 96K largest component) and **4.6s average verification time** per sample at practical ε are achieved through compositional verification. New experts integrate by verifying only the new component—existing verifications remain valid. **16.7% training efficiency improvement** is achieved for 3-expert systems via frozen-expert router fine-tuning.

5. Discussion and Limitations

When Does Compositional Verification Apply? The framework requires router stability within $B_\varepsilon(x)$. For dissimilar domains (CIFAR-10 vs MNIST), routers learn large decision margins enabling easy certification. For similar domains (e.g., GTSRB vs PTSD traffic signs), routing accuracy degrades to 93%, and compositional guarantees degrade unless the router achieves sufficiently large margins; this may require stronger adversarial training or probabilistic certification.

Architecture Scope. Our verification-compatible CNNs (96K parameters) demonstrate the framework but limit expressiveness. Scaling to Transformers faces verifier memory constraints. Promising directions include verification-aware architecture search and combining modular verification with randomized smoothing for probabilistic certificates on larger models.

Relationship to Empirical Robustness. Adversarial training improves certifiability by smoothing decision boundaries, reducing bound looseness in CROWN analysis. This explains why robust training simultaneously improves empirical adversarial accuracy and formal certification rates.

6. Conclusion

We establish that heterogeneous MoE robustness composes modularly under top-1 routing: verified router stability combined with verified expert robustness yields verified system robustness.

Our experiments reveal **13.1 percentage point** accuracy improvement from expert robustness, which dominates system behavior, while router training is negligible for dissimilar domains (<0.1%). Critically, robust training is a prerequisite for formal verifiability—non-robust components produce decision boundaries incompatible with sound verification.

These results demonstrate that compositional design enables tractable formal verification of modular AI systems. Future work includes extending guarantees to top-k routing and scaling to transformer-based architectures.

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A. Proof of Theorem 3.1

Proof. We prove that under the stated conditions, the MoE system is ε-robust at input x with label y.

Setup. Let $\mathrm{MoE} = (\mathcal{G}, \{E_i\}_{i=1}^N)$ be a heterogeneous mixture-of-experts with top-1 routing. For input x, let $m^* = \arg\max_i \mathcal{G}_i(x)$ denote the selected expert. The MoE prediction is defined as:

$$\mathrm{MoE}(x) = E_{m^*}(x)$$

Goal. Show that $\forall x' \in B_\varepsilon(x) : \arg\max \mathrm{MoE}(x') = y$.

Proof. Let $x' \in B_\varepsilon(x)$ be an arbitrary perturbation satisfying $\|x' - x\|_\infty \leq \varepsilon$.

Step 1 (Router Stability). By assumption (1):

$$\forall x' \in B_\varepsilon(x) : \arg\max_i \mathcal{G}_i(x') = m^*$$

Thus, the router selects expert E_{m^*} for all perturbations in $B_\varepsilon(x)$. This implies:

$$\mathrm{MoE}(x') = E_{m^*}(x') \quad \forall x' \in B_\varepsilon(x)$$

Step 2 (Expert Robustness). By assumption (2):

$$\forall x' \in B_\varepsilon(x) : \arg\max E_{m^*}(x') = y$$

The selected expert correctly classifies all perturbations within the ε-ball.

Step 3 (Composition). Combining Steps 1 and 2:

$$\arg\max \mathrm{MoE}(x') = \arg\max E_{m^*}(x') = y \quad \forall x' \in B_\varepsilon(x)$$

Therefore, the MoE system maintains correct classification for all inputs in $B_\varepsilon(x)$, satisfying the definition of ε-robustness. □

B. Architecture and Training

VerifiableCNN (96K parameters): 4 convolutional blocks ($3 \to 20 \to 28 \to 40 \to 56$ channels), 3×3 kernels, ReLU activations, average pooling. Global average pooling $\to$ FC(56, 128) $\to$ ReLU $\to$ FC(128, C). No BatchNorm layers (folded into convolutional weights for ONNX export).

Training: Adam optimizer, learning rate 10^{-3}, batch size 128. Experts: 200 epochs; Router: 50 epochs on frozen experts. RT variants: PGD adversarial training ($\varepsilon = 8/255$, 7 steps, step size $2/255$).

Hardware: Intel i7-14700K CPU, NVIDIA RTX 4060 8GB GPU, 64GB RAM. PyTorch 2.1, α,β-CROWN (December 2024 release), Gurobi 11.0 LP solver. Verification timeout: 300 seconds per sample.

C. Similar Domain Experiments

GTSRB + PTSD (both traffic sign datasets): routing accuracy degrades to 93.17% (vs 100% for CIFAR-10 + MNIST), adversarial accuracy drops to 29.47% (vs 41.59%). The 6.83% routing error propagates to 12.12% system accuracy reduction, demonstrating that compositional guarantees depend on router margin geometry.